Cardiorespiratory Monitoring using RF, Lidar, and Camera: A Comparison with Synchronized Multimodal Measurements

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Abstract

While the viability of radio-frequency (RF), camera, and lidar systems for contactless cardiorespiratory monitoring has been demonstrated, a common platform for objective comparison across the three modalities is still lacking. To that end, this paper presents comprehensive, synchronized multimodal measurements collected with our 28 GHz context-aware channel sounder, combining RF heatmaps, lidar point clouds, and camera RGB images with physiological reference sensors; the measurements span multiple subjects, postures, body orientations, and breathing regimes, totaling 189 cases and more than 90 000 RF heatmaps, 23 000 point clouds, and 16 000 images. Of note, we also propose a novel, off-the-shelf respiration estimation method for the camera—the lowest-cost and most pervasive of the three systems—that uses an AI-based general-purpose monocular relative-depth estimator; by aggregating thoracic relative depth over time, the method recovers a breathing waveform from weak out-of-plane chest motion. Finally, we compare both respiration rate (RR) and heart rate (HR) estimation across the three modalities. The results show that all three detect RR, with the camera achieving the highest overall detection probability, while revealing modality-specific errors; for HR, RF achieves the highest detection probability, whereas camera yields the most precise estimates when the optical pulse signal is visible. Beyond single-modality comparison, the synchronized measurements provides a benchmark for exploring algorithms that fuse RF, camera, and lidar streams to optimize performance.

1 Introduction

Integrated sensing and communication (ISAC) envisions next generation wireless networks that infer the state of the surrounding environment. In its broader multimodal form, multimodal ISAC (M-ISAC) extends radio sensing with complementary sensors such as cameras and lidar, so that perception is not limited to a single physical modality [3]. This extension is important when radio measurements are informative but incomplete. RF sensing captures subtle motion and operates under conditions where optical sensing is fragile, while vision and depth measurements provide geometric and appearance cues that are easier to interpret. The value of M-ISAC lies in the utilization of complementary sensing mechanisms.

In this context, contactless respiration monitoring is a relevant use case because breathing carries information about physiological state, fatigue, and cardiopulmonary health. This interest has produced a broad range of sensing approaches. Radio Frequency (RF) infers chest motion from phase, range, or Doppler variations [2]. Cameras estimate vital signs from subtle appearance changes or body motion [1, 4]. Depth sensors recover small thoracic displacements directly in three dimensions [16]. Wearable devices provide a practical reference for validation. These studies show that respiration rate (RR) and heart rate (HR) can be measured without direct contact using physical modalities.

The main challenge is to compare these modalities under matched experimental conditions. Several public datasets address parts of this problem. The Radar Human Breathing Dataset combines radar and lidar for respiratory monitoring [8]. Other resources combine RF sensing with visual measurements and respiratory references [5]. Fusion-Vital provides synchronized RGB video and RF reflections for respiratory and cardiac signal estimation [6]. OMuSense-23 combines mmWave radar and RGB-D data for breathing-related activities [11], while PhysDrive provides synchronized RGB, near-infrared camera, and raw mmWave radar data for in-vehicle physiological monitoring [14]. These datasets confirm the importance of synchronized multimodal data, but do not provide synchronized RF, camera, and lidar measurements with common RR and HR references for a fair comparison.

This paper addresses this gap with a synchronized multimodal dataset that aligns RF heatmaps, RGB video, lidar point clouds (PCs), and vital-sign references. Measurement data presented in this paper are available at [12]. The campaign records all modalities during the same sessions, with shared timing, common geometry, and controlled variations in posture, body orientation, breathing regime, and single- or multiple-person occupancy. In addition, we define a common evaluation procedure for RR and HR estimation across RF, camera, and lidar. Each modality is processed independently and compared against the same wearable reference under matched conditions. Beyond the dataset, our second contribution focuses on the camera system, the least expensive and most pervasive modality. Unlike prior work that uses monocular depth together with optical flow as inputs to a respiration-specific learning model [10], our approach uses a general-purpose AI-based monocular relative-depth estimator directly as the respiratory observable. The resulting method is off-the-shelf, requires no respiration-specific training, and recovers breathing by aggregating thoracic depth variations from RGB video. This directly targets the weak out-of-plane chest displacement produced by breathing, extracting a geometric cue rather than relying only on RGB intensity variation or two-dimensional image motion, while yielding a camera-only, non-contact, and readily deployable implementation. In summary, this paper makes three main contributions:

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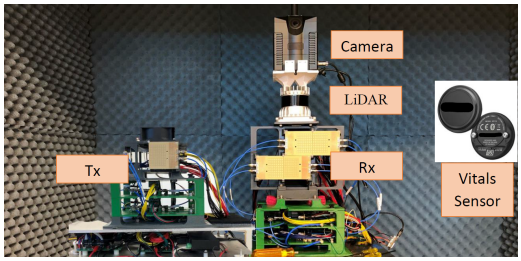
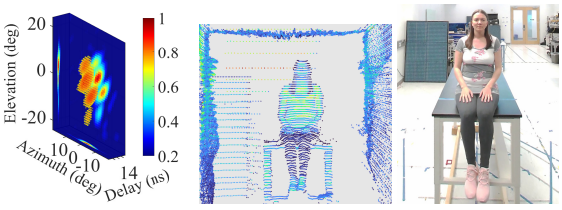


Figure 1: 28 GHz context-aware channel sounder



(a) RF heatmap (b) Lidar output (c) Camera output

Figure 2: Representative modality outputs

- *Synchronized multimodal measurements:* we release a synchronized dataset for contactless cardiorespiratory monitoring that combines 28 GHz RF heatmaps, RGB video, PCs, and wearable RR/HR references under common subjects, geometry, and timing.
- *AI-assisted camera-based respiration estimation:* we propose a novel, off-the-shelf method for the camera, the lowest-cost and most pervasive modality, that exploits AI-based monocular relative depth to convert thoracic out-of-plane motion directly into a breathing waveform.
- *Objective cross-modality comparison:* we evaluate RF, camera, and lidar RR/HR estimation enabling a measurement-based analysis of the accuracy, failure modes, and complementarity of the three modalities.

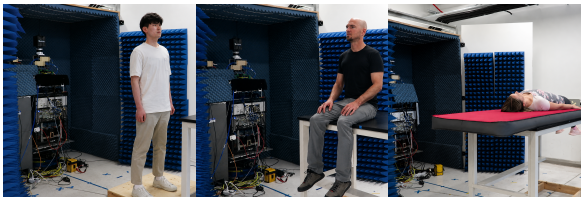
The remainder of the paper is organized as follows. Section II describes the multimodal sounder and measurements campaign. Section III presents the vital-sign extraction for each sensor. Section IV reports the RR and HR estimation results. Section V concludes the paper.

2 Channel Sounder and Measurements

The measurements were acquired with our 28 GHz context-aware channel sounder, integrating lidar, camera, and wearable reference sensors, as shown in Fig. 1. The full hardware architecture, calibration methodologies, and data processing are described in [7, 9].

2.1 Multimodal Channel Sounder

The RF subsystem is a 28 GHz channel sounder with co-located transmitter and receiver, so the subject is observed from a common viewpoint. The transmitter uses an 8×8 phased-array front end to illuminate the scene, while the receiver combines four 8×8 phased-array boards into an extended 256-element aperture. Matched filtering and digital beam synthesis convert the measured array responses into RF heatmaps indexed by delay, azimuth angle-of-arrival, elevation angle-of-arrival, and time; each heatmap cell retains the associated path gain and phase, as shown in Fig. 2a.



(a) Standing (b) Sitting (c) Lying

Figure 3: Body positions used during the measurements

The optical sensing stack provides spatial context for the same scene. A mechanical 360° lidar scanner mounted at the receiver captures a $360° \times 90°$ field of view with 0.17° azimuth resolution, 0.7° elevation resolution, and 1 cm range resolution at 100 ms frame intervals. The resulting PCs represent the subject and surrounding scene in absolute three-dimensional coordinates, as shown in Fig. 2b. A RGB camera mounted above the lidar records $360° \times 145°$ spherical images, as shown in Fig. 2c, with a resolution of 11000×5500 pixels every 140 ms.

Wearable sensors provide the physiological reference. The electrocardiogram (ECG) signal is used as the HR reference, while the inertial measurement unit (IMU) records three-axis acceleration and gyroscope data that capture RR. In multi-subject measurements, each subject wears a separate sensor so that the estimated vital signs can be compared against subject-specific ground truth (GT).

The dataset enables direct comparison across modalities by aligning RF, lidar, camera, and wearable measurements on a common spatial and temporal reference. The RF, lidar, and camera sensors are co-mounted at the receiver, and metallic calibration spheres visible across modalities are used to correct residual offsets and align the optical data with the RF heatmap coordinates. The RF digitizer and lidar share a rubidium-disciplined clock and hardware trigger, while the camera is aligned offline by correlating motion events with the lidar stream. A manual tap on the wearable sensor at the beginning of each session provides a synchronization marker for the ECG and IMU streams. For each case, synchronization tables map every RF heatmap index to the corresponding camera image, lidar frame, ECG sample, and IMU sample. The RF sequence is used as the common time axis, with the camera and lidar streams mapped onto it.

2.2 Measurement Campaign

The measurement campaign was designed to capture RR under controlled changes in posture, orientation, and occupancy. The dataset contains measurements from three individual subjects and three pairwise subject combinations. The single-subject portion spans three body postures, namely standing, sitting, and lying down, and three orientations with respect to the sensing assembly, namely front, side, and back. The pairwise portion measures two people simultaneously in the same three body postures, and in the same three orientations. Figure 3 shows the three body positions used in the single-subject recordings. Each subject is instructed to follow a predefined breathing pattern during the acquisition. The protocol includes normal breathing for 25 s, breath holding for 10 s, and fast breathing for 15 s. These intervals are recorded sequentially, capturing transient respiration regimes and enabling the benchmark to evaluate both RR/HR estimation accuracy and the response of each modality to abrupt changes in respiratory activity.

In total, the benchmark contains 189 cases for a total of 39 minutes of recording, 90 000 RF heatmaps, more than 16 000 images and more than 23 000 PCs.

3 Cardiorespiratory Estimation Methods

This section describes single-modality physiological signal estimation from camera, RF, and lidar measurements, beginning with our proposed respiration estimation method based on AI-assisted camera monocular relative depth. Each algorithm provides a one-dimensional signal, which is then processed with the same spectral estimator for RR and HR estimation.

3.1 Proposed: Camera Physiological Signal

The camera system estimates RR and HR from the RGB sequence using two complementary image-derived traces. For RR, we introduce an AI-assisted monocular-depth approach that uses relative depth in the thoracic region rather than RGB intensity variation or two-dimensional pixel motion. This is a key contribution of the camera modality. Since breathing mainly induces weak out-of-plane chest displacement, the depth representation provides a direct geometric cue for the underlying physiological motion, while reducing sensitivity to clothing texture, illumination changes, and small apparent RGB shifts. The method therefore turns a standard RGB camera into a depth respiratory sensor without using a physical depth camera. The approach remains camera-only and non-contact, and it avoids respiration-specific training by relying on a pretrained monocular depth estimator. As a result, the proposed camera method exploits modern AI-based visual depth estimation while remaining camera-only, training-free, and easy to reproduce.

A YOLO-based person detector [13] is first applied to each frame to localize the subject with a bounding box. The detector is used only to define the subject region of interest over time. A monocular depth estimator based on Depth Anything [15] is then applied to each RGB frame to obtain a dense relative depth map. Depth Anything uses teacher–student training with pseudo-depth labels from unlabeled images, strong data augmentation, and pretrained visual encoders to improve zero-shot generalization. Let $D(x, y, t_n)$ denote the relative depth at pixel (x, y) and frame index n. For each frame, the person bounding box defines a chest-centered region of interest Ω_n. The camera respiratory signal is computed as:

$$s_{\text{cam}}^{(\text{RR})}(t_n) = \frac{1}{|\Omega_n|} \sum_{(x,y)\in\Omega_n} D(x, y, t_n), \tag{1}$$

where $|\Omega_n|$ is the number of pixels in the selected region. This signal represents the temporal evolution of thoracic displacement inferred from the camera.

The same RGB sequence is used to estimate HR with a remote photoplethysmography (rPPG) baseline. Although rPPG methods include blind-source-separation, chrominance-based, projection-based, and learning-based approaches [4], this work uses the green-channel method because it is training-free and reproducible. Let $G(x, y, n)$ denote the green-channel intensity and let Φ_n denote a face or skin region of interest. The camera HR signal is:

$$s_{\text{cam}}^{(\text{HR})}(t_n) = \frac{1}{|\Phi_n|} \sum_{(x,y)\in\Phi_n} G(x, y, t_n), \tag{2}$$

where $|\Phi_n|$ is the number of pixels in the selected region. HR is estimated from the dominant frequency of $s_{\text{cam}}^{(\text{HR})}(t_n)$ within the expected cardiac band.

3.2 RF Physiological Signal

The RF baseline extracts a physiological trace from complex delay-angle RF heatmaps. Let $H(\tau, \theta, t_n)$ denote the channel response at delay bin τ, angular bin θ, and frame index t_n. For each angle, a sliding-window short-time Fourier transform (STFT) is applied along slow time to obtain the block-wise range-Doppler representation $R_\theta(f_D, t_{n'}) = |\text{STFT}\{H(\tau, \theta, t_n)\}|$, where f_D is the Doppler frequency and $t_{n'}$ indexes the slow-time block. A CFAR filter is applied to each block to obtain $\hat{R}_\theta(f_D, t_{n'})$, which suppresses weak components. The micro-Doppler map is then formed by aggregating the thresholded range-Doppler responses across delay and angle. The RF trace s_{RF}, describing the temporal evolution of the chest radial velocity, is extracted by tracking the dominant velocity ridge across slow-time blocks as:

$$s_{\text{RF}}(t_{n'}) = \frac{\lambda}{2} \arg\max_{f_D\in\mathcal{B}_v} \left[\sum_{\theta\in\Theta} \sum_{\tau\in\mathcal{T}} \hat{R}_\theta(f_D, t_{n'}) \right]. \tag{3}$$

3.3 Lidar Physiological Signal

The lidar system estimates respiratory motion from the temporal evolution of depth in the thoracic region. For each point of the PC, indexed by coordinates (i, j), the distance from the sensor is computed directly from the three-dimensional coordinates as $D(i, j) = \sqrt{x_{ij}^2 + y_{ij}^2 + z_{ij}^2}$, where (x_{ij}, y_{ij}, z_{ij}) denotes the Cartesian position of the point. This step converts the PC into a depth map.

The resulting sequence of depth maps is then processed over time. A region of interest is selected around the subject's torso, and the same region is tracked across consecutive frames. Let $D(i, j, t_n)$ denote the depth value at pixel (i, j) and frame index t_n. Denoting the tracked torso region by Ω_n, the lidar physiological trace is obtained by averaging the depth values within the region of interest,

$$s_{\text{lidar}}(t_n) = \frac{1}{|\Omega_n|} \sum_{(i,j)\in\Omega_n} D(i, j, t_n), \tag{4}$$

where $|\Omega_n|$ is the number of pixels in the selected region. The signal $s_{\text{lidar}}(t_n)$ represents the temporal evolution of the chest displacement measured directly from lidar depth.

3.4 Breathing- and Heart-Rate Estimation

The final stage is the same for RF, camera, and lidar. The physiological signals s_{RF}, $s_{\text{cam}}^{\text{RR}}$,$s_{\text{cam}}^{\text{HR}}$, or s_{lidar} carry the same temporal periodicity. The depth signal from camera or lidar contains the breathing and heartbeat frequencies directly through the motion of the chest or the variation of the pulse. The radial-velocity signal from RF micro-Doppler contains the same frequencies because it represents the time derivative of that motion.

Each physiological signal is first detrended and then separated into two frequency ranges: an RR band from 0.1 Hz to 0.8 Hz and an HR band from 0.8 Hz to 2.0 Hz. For each band, the power spectral density is computed using Welch's method. The RR is estimated as the dominant spectral peak in the RR band, while the HR is estimated as the dominant spectral peak in the HR band. A detection is declared if the peak signal-to-noise ratio exceeds 3 dB.

4 Experimental Results

This section evaluates RR and HR detection across RF, camera, and lidar, and uses the synchronized measurements to cross-validate trends across modalities.

4.1 Qualitative Results

This subsection illustrates representative time-domain signals s_{RF}, $s_{\mathrm{cam}}^{\mathrm{RR}}$, $s_{\mathrm{cam}}^{\mathrm{HR}}$, and s_{lidar}, together with their spectra, extracted from the RF, camera, and lidar measurements. Each example compares the dominant spectral peak estimated from the sensing modality with the reference, namely the IMU for RR and the ECG for HR.

Figure 4 shows a single-person standing case with the chest facing the sensors. In this case all modalities produce consistent respiratory estimates. The time-domain signals exhibit a clear periodic component, and the corresponding spectra show a dominant peak aligned with the IMU reference. This example confirms that RF, camera, and lidar can recover the fundamental breathing frequency when the respiratory motion is visible in each sensing mode.

Figure 5 shows a heartbeat example for the pairwise sitting-front case. Compared with respiration, the heartbeat signals are noisier because they are weaker and easily masked by body motion, illumination changes, and noise. In this case, both the RF and camera spectra contain energy near the ECG reference. However, the largest peak in the camera spectrum occurs at a different frequency, which leads to an incorrect estimate. The performance of the rPPG-based camera detector depends on how light reaches the visible skin region. When illumination changes corrupt the color trace, or when visible skin is limited by hair and clothing, the strongest spectral component may no longer correspond to the heartbeat.

Figure 6 shows a pairwise standing-back case. In this configuration, the camera and lidar estimates remain close to the IMU reference, while the RF estimate is biased toward a higher frequency. The RF signal appears more sensitive to small body motions that occur while the subject is standing. A weak spectral shoulder near the expected breathing frequency suggests that the correct component may still be present, but it is not sufficiently resolved by the adopted spectral estimator.

These examples show that the sensing modalities are complementary. A modality that performs well in one posture, orientation, or vital-sign regime may fail in another, while a different modality may remain reliable.

4.2 Respiration-Rate Estimation

All three modalities detect breathing in most trials with a high probability of detection (P_{d}). Table 1 reports $P_{\mathrm{d}} = 0.810$ for camera, $P_{\mathrm{d}} = 0.802$ for lidar, and $P_{\mathrm{d}} = 0.762$ for RF. These values show that RR estimation is feasible with each modality.

The error distributions in Fig. 7a show different error structures after detection. Lidar has the most compact distribution around zero. Camera remains close to zero but has a slightly broader spread. RF also centers near zero, but its distribution is wider and has longer tails. This behavior is consistent with the physical quantity measured by each sensor. Camera and lidar observe visible motion or depth changes, whereas RF senses radial micro-motion through phase and Doppler variations in a multipath channel. RF can therefore capture small respiratory motion, but body movements can also induce phase fluctuations and Doppler components that are not directly related to chest displacement.

Table 1: Probability of detection P_{d} for RR estimation.

Condition	Case	N_{cases}	Camera	Lidar	RF
Overall	All	126	0.810	0.802	0.762
Posture	Lying	18	0.833	0.722	0.722
	Sitting	54	0.815	0.833	0.778
	Standing	54	0.796	0.796	0.759
Body orient.	Back	42	0.857	0.810	0.786
	Front	42	0.810	0.857	0.810
	Side	42	0.762	0.738	0.690
Mode	Fast	63	0.921	0.952	0.698
	Normal	63	0.937	0.889	0.825
Subject	Subject 1	42	0.881	0.905	0.619
	Subject 2	42	1.000	0.952	0.905
	Subject 3	42	0.857	0.857	0.762
Setting	Single subject	54	0.815	0.759	0.741
	Multiple subjects	72	0.806	0.833	0.778

Posture changes the error distribution more than it changes the P_{d}. Table 1 shows comparable P_{d} values across lying, sitting, and standing for the three modalities. Camera ranges from 0.796 to 0.833, lidar from 0.722 to 0.833, and RF from 0.722 to 0.778. Fig. 7b shows a stronger posture effect on RF error spread. RF is more compact in lying and sitting trials than in standing trials. This trend is consistent with reduced body motion. Sitting also provides a favorable RF view of the chest, while lying can reduce the direct reflected component from the chest and make the estimate rely more on scattered energy, as shown in Figure 3. Camera and lidar remain more compact across postures, which suggests that their visible-motion and depth measurements are less affected by this posture change.

Body orientation produces a trend shared by all three modalities. Side orientation gives the lowest P_{d} for camera, lidar, and RF. Table 1 reports $P_{\mathrm{d}} = 0.762$ for camera, 0.738 for lidar, and 0.690 for RF in the side view. Front and back views remain higher for all three sensors. For RF, the side view reduces the radial component of respiratory motion. For camera and lidar, the chest is less directly visible, and the projected thoracic area is smaller. The breathing signal therefore becomes weaker for all modalities. The error distributions in Fig. 7c show that orientation affects RF more strongly than camera and lidar. RF has a wider spread in the back view, while the front and side views remain more centered around zero.

Breathing mode affects P_{d} differently across modalities. Table 1 shows that camera and lidar maintain a high P_{d} in both breathing modes, whereas RF decreases from 0.825 to 0.698, when the subject breathes fast. These results suggest that camera and lidar are more robust to fast breathing, while RF is more sensitive to the additional body motion generated by fast breathing, which induces phase variations in the received signal. The error distributions in Fig. 7d show that fast breathing produces compact errors after successful detection. This behavior indicates that fast breathing can generate a strong periodic component if subjects remain sufficiently still.

Subject identity affects detection probability in a similar way across the modalities. Table 1 shows that Subject 1 has the lowest P_{d}, Subject 2 has the highest P_{d}, and Subject 3 lies between them. This pattern appears for camera, lidar, and RF, which suggests that subject-specific breathing motion or body signature affects all

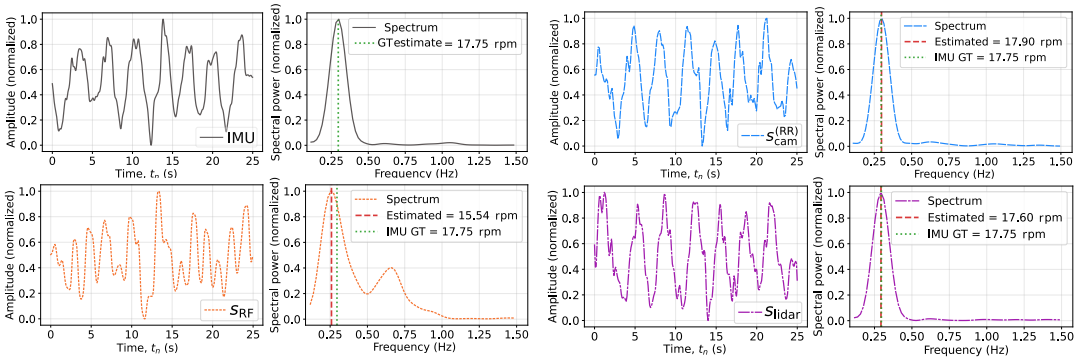


Figure 4: Respiratory signals for the single-person standing case using four sensing modalities: IMU GT, $s_{cam}^{(RR)}$, s_{RF}, and s_{lidar}.

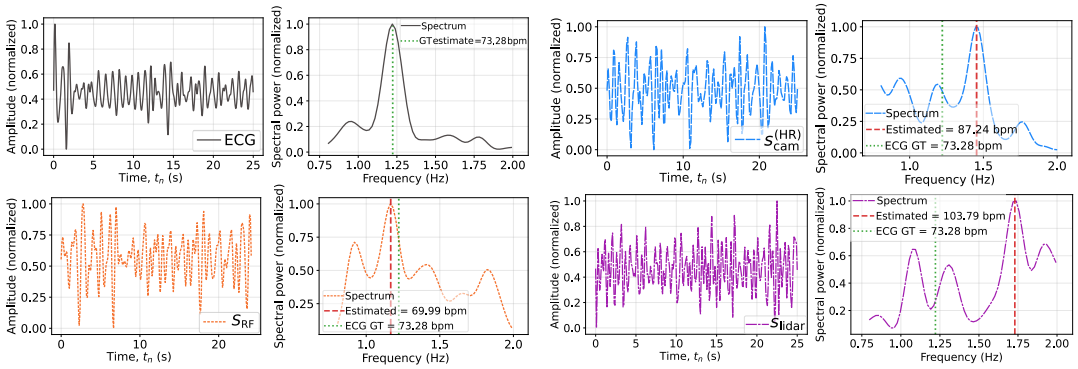


Figure 5: Heartbeat signals for the pairwise sitting-front case using four sensing modalities: ECG GT, $s_{cam}^{(HR)}$, s_{RF}, and s_{lidar}.

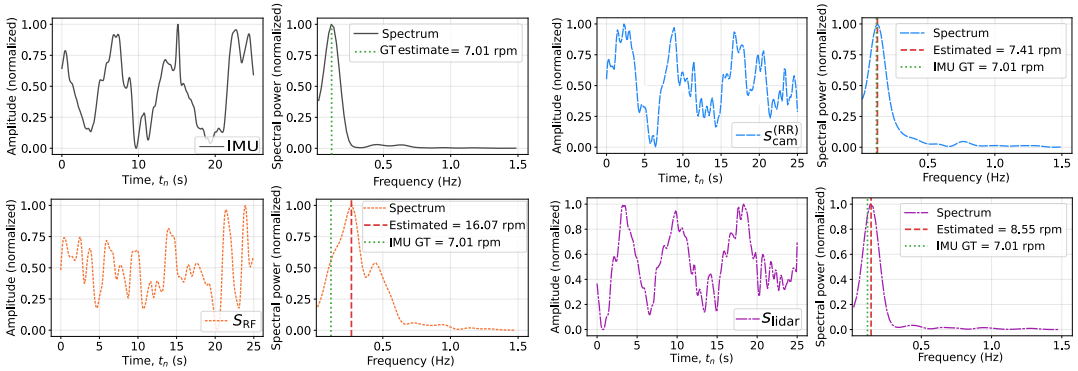


Figure 6: Respiratory signals for the pairwise standing-back case using four sensing modalities: IMU GT, $s_{cam}^{(RR)}$, s_{RF}, and s_{lidar}.

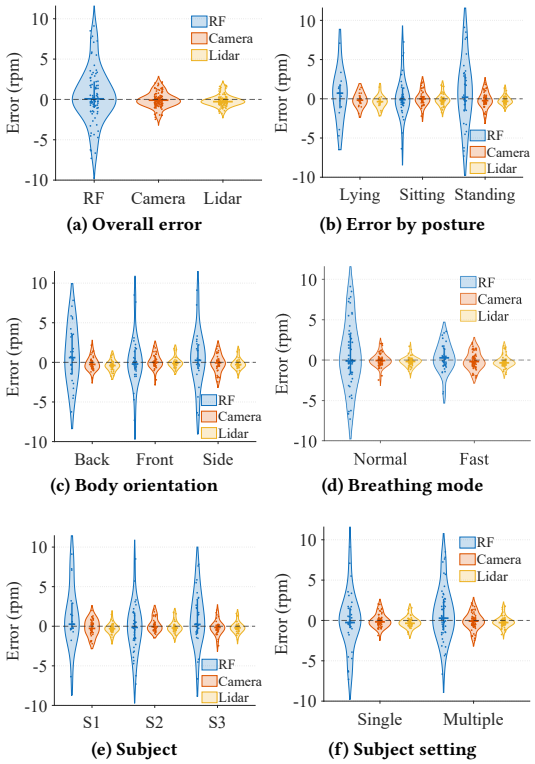
modalities. Fig. 7e shows that the error distributions remain broadly similar across subjects after detection. Camera and lidar remain compact around zero. RF keeps a wider distribution for all subjects.

The side-by-side multiple-subject setting does not cause a clear degradation in RR estimation. Table 1 reports similar P_d values for single-subject and multiple-subject trials. Camera changes from 0.815 to 0.806, lidar changes from 0.759 to 0.833, and RF changes from 0.741 to 0.778. Fig. 7f supports the same interpretation.

4.3 Heart-Rate Estimation

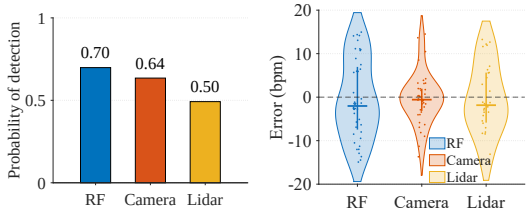
HR detection is more difficult than RR detection, but the synchronized measurements show complementary behavior across modalities. The P_d is 0.635 for camera, 0.492 for lidar, and 0.699 for RF. RF gives the highest detection probability, which is consistent with its sensitivity to small periodic motions produced by cardiac activity.

The error distributions in Fig. 8 show different post-detection error structures. RF and lidar exhibit broad bell-shaped distributions, indicating that detected HR peaks can deviate substantially from the wearable reference. Camera shows a more compact distribution around zero. This behavior is consistent with the sensing principle used by the camera: the camera estimate relies on pulse-related color changes in exposed skin. This pulse signal is more likely to be available in front-facing configurations, where most of the facial skin is visible to the camera. In side- and back-facing configurations, the available skin area can decrease substantially, and additional



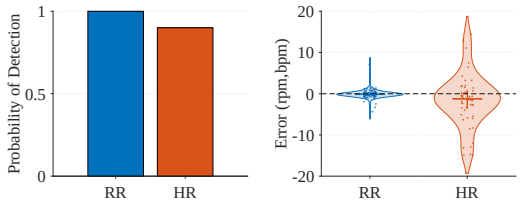
(a) Overall error (b) Error by posture
(c) Body orientation (d) Breathing mode
(e) Subject (f) Subject setting

Figure 7: Error swarm and violin plots for RR estimation.



(a) Probability of detection (b) Overall error

Figure 8: P_d, error swarm and violin plots for HR estimation



(a) Probability of detection (b) Overall error

Figure 9: Probability of detection, error swarm, and violin plots for multimodal oracle estimator

factors such as hairstyle, hair coverage, and head orientation can reduce the amount of illuminated skin observed by the camera.

4.4 Oracle Multimodal Estimator

We finally prove the value of M-ISAC. For each recording, an oracle chooses the RF, camera, or lidar estimate with the smallest absolute error relative to the wearable reference. Fig. 9 shows an improvement over fixed single-sensor estimation. The oracle reaches $P_{\mathrm{d}} = 1.00$ for RR and $P_{\mathrm{d}} = 0.90$ for HR. The error distributions also concentrate more tightly around zero, showing that multimodal processing suppresses many large errors produced by individual modalities. This result shows the promise of M-ISAC and demonstrates that the dataset exposes cross-modality complementarity, motivating algorithms that jointly exploit multiple sensing modalities rather than relying on a single sensor.

5 Conclusion

This paper presented a synchronized RF, camera, and lidar dataset for contactless cardiorespiratory monitoring with common subjects, geometry, timing, and wearable references. The results show that all three modalities estimate RR, while HR estimation remains more challenging and exposes modality-specific errors. Camera and lidar provide compact RR errors, RF achieves the highest HR detection probability, and the oracle multimodal result shows that selecting across modalities improves both RR and HR estimation. These findings support M-ISAC as a practical direction for robust contactless vital-sign sensing and provide a benchmark for novel multimodal algorithms.

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